

Parabolic Tunneling Calculations

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A new analytic approximation is presented for thermally averaged transmission coefficients for tunneling through and reflection by truncated parabolic potential barriers. The approximation has a wide range of validity and avoids the spurious singularities of the well-known result for untruncated parabolic barriers. We also apply the analytic result to effective parabolic fits to other barrier shapes. The results are quite encouraging and suggest that the approximation may be useful for a wide variety of chemical applications.

I. Introduction

Tunneling through potential energy barriers occurs in many contexts in chemistry. The simplest realistic approximation to a potential energy barrier is a truncated parabola. Thermally averaged transmission coefficients for truncated parabolas have usually been approximated by the method of Wigner¹ or that of Bell.² In this article we present a more satisfactory solution to this problem. We use this solution to calculate approximate transmission coefficients for more general barrier shapes and compare the results to accurate uniform semiclassical (i.e., uniformized WKB) or quantal calculations.

II. Theory

A. Derivation. Symmetric Barrier. The problem under consideration is transmission through a one-dimensional potential barrier. First consider a symmetric barrier

$$V(x) = V_0 - \frac{1}{2}m\omega^2x^2 \quad |x| \leq (2V_0/m\omega^2)^{1/2} \\ V = 0 \quad |x| > (2V_0/m\omega^2)^{1/2} \quad (1)$$

where m is the particle mass and V_0 and ω are parameters of the potential. The object is to calculate a good approximation to the transmission coefficient at temperature T . This is defined for an arbitrary barrier as³

$$\kappa(T) = \frac{\int_0^\infty T(E) \exp(-\beta E) dE}{\int_0^\infty T_C(E) \exp(-\beta E) dE} \quad (2)$$

where $\beta = 1/k_B T$, k_B is Boltzmann's constant, and $T(E)$ and $T_C(E)$ are the quantum mechanical and classical transmission coefficients at energy E . From (1) we have

$$T_C(E) = \theta(E - V_0) \quad (3)$$

where $\theta(y)$ is a step function increasing from 0 to 1 at $y = 0$. Putting (3) into (2) yields

$$\kappa(T) = \beta \exp(\beta V_0) I \quad (4)$$

where

$$I = \int_0^\infty T(E) \exp(-\beta E) dE \quad (5)$$

We approximate $T(E)$ by the uniformized WKB result for a parabola⁴

$$T(E) = \frac{1}{1 + \exp[\alpha(V_0 - E)]} \quad (6)$$

where

$$\alpha = 2\pi/(\hbar\omega)$$

It is convenient to divide the integral into two terms

$$I = \tilde{I}(0, V_0) + \tilde{I}(V_0, \infty) \quad (7)$$

where

$$\tilde{I}(E_1, E_2) = \int_{E_1}^{E_2} \frac{\exp(-\beta E)}{1 + \exp[\alpha(V_0 - E)]} dE \quad (8)$$

Consider first $\tilde{I}(0, V_0)$. Here $E < V_0$, so we expand the integrand as follows

$$\tilde{I}(0, V_0) = \int_0^{V_0} \exp(-\beta E) \sum_{n=0}^{\infty} (-1)^n \exp[-(n+1)\alpha(V_0 - E)] dE \quad (9)$$

For the second term, $V_0 < E$, and we choose a different expansion

$$\tilde{I}(V_0, \infty) = \int_{V_0}^{\infty} \exp(-\beta E) \sum_{n=0}^{\infty} (-1)^n \exp[n\alpha(V_0 - E)] dE \quad (10)$$

Integrating (9) and (10) and substituting into (7) and (4) yields

$$\kappa(T) = \sum_{n=0}^{\infty} (-1)^n \beta \left(\frac{1 - \exp[(\beta - (n+1)\alpha)V_0]}{(n+1)\alpha - \beta} + \frac{1}{n\alpha + \beta} \right) \quad (11)$$

When $\beta < \alpha$, the nonexponential parts of the sums can be combined and we get

$$\kappa(T) = \frac{\pi\beta/\alpha}{\sin(\pi\beta/\alpha)} - \sum_{n=0}^{\infty} \frac{(-1)^n \beta \exp[(\beta - (n+1)\alpha)V_0]}{(n+1)\alpha - \beta} \quad (\beta < \alpha) \quad (12)$$

As β approaches α from below, the pole in the first term of (12) is cancelled by the $n = 0$ term in the sum. When α is large the $n = 0$ term dominates the sum, and

$$\kappa(T) \simeq \frac{\pi\beta/\alpha}{\sin(\pi\beta/\alpha)} + \frac{\beta}{\beta - \alpha} \exp[(\beta - \alpha)V_0] \quad (\alpha V_0 \gg 1) \quad (13)$$

For the case $\alpha = \beta$ this yields

(1) E. Wigner, *Z. Phys. Chem. B*, **19**, 203 (1932).

(2) R. P. Bell, *Trans. Faraday Soc.*, **55**, 1 (1959). Note that our definitions of α and β are different from Bell's.

(3) See, e.g., D. G. Truhlar and A. Kuppermann, *J. Am. Chem. Soc.*, **93**, 1840 (1971).

(4) E. C. Kemble, "The Fundamental Principles of Quantum Mechanics with Elementary Applications", McGraw-Hill, New York, 1937, pp 109ff.

$$\kappa(T) = \beta V_0 = \alpha V_0 \quad (\alpha = \beta) \quad (14)$$

When $\alpha < \beta$, the combination of series to yield (12) is not valid. However, for large β one can show that²

$$\kappa(T) \simeq \frac{\beta}{\beta - \alpha} \exp[(\beta - \alpha)V_0] \quad (\beta V_0 \gg 1) \quad (15)$$

By retaining the next largest term in (11) we get

$$\kappa(T) \simeq \frac{\beta}{\beta - \alpha} \{\exp[(\beta - \alpha)V_0] - 1\} \quad (\beta \geq \alpha) \quad (16)$$

Equation 16, like (13), yields the limit (14). Thus eq 13 and 16 provide a divergence-free approximation to $\kappa(T)$ that is a continuous function of α and β for all possible values of these parameters.

B. Comparison to Previous Work. Wigner's tunneling correction is valid for $\beta \ll \alpha$. It can be obtained by expanding eq (13):

$$\kappa(T) \simeq 1 + (\pi^2 \beta^2 / 6 \alpha^2) \quad (\beta \ll \alpha) \quad (17a)$$

which is usually written in the form

$$\kappa(T) \simeq 1 + (\hbar^2 \omega^2 \beta^2 / 24) \quad (\beta < \alpha) \quad (17b)$$

Bell² discussed the three limiting forms, eq 12, 14, and 15. In the present work, by retaining the extra terms in (13) and (16) we obtain a continuous solution that connects these limits in a divergence-free manner. In section III we will show that eq 13 and 16 also provide useful numerical accuracy.

It is also interesting to compare the present treatment to the well-known result⁵ for an infinite (i.e., untruncated) parabolic barrier:

$$\kappa(T) = \frac{1/2 \beta \hbar \omega}{\sin(1/2 \beta \hbar \omega)} \quad (18)$$

This result shows singularities whenever $1/2 \beta \hbar \omega$ is an integral multiple of π , i.e., at $\beta = (j + 1)\alpha$, $j = 0, 1, 2, \dots$. The equations in section A show that these are a result of the artificial nature of the untruncated parabola. For example, with $n = 0$, the correct result (14) does not diverge. Consider next the divergent terms with $n = 1$ in eq 11. The limiting behavior of these terms is

$$\lim_{\beta \rightarrow 2\alpha} \left(-\frac{\beta}{2\alpha - \beta} \{1 - \exp[(\beta - 2\alpha)V_0]\} \right) = \beta V_0 \quad (19)$$

Thus there is no divergence. Furthermore, for $\beta = 2\alpha$, the $n = 0$ term already contributes $\exp(\beta V_0/2)$ to $\kappa(T)$. For sufficiently large α and β , this dominates, i.e., $\exp(\alpha V_0) > 2\alpha V_0$, as already implied by eq 15. Although it has been popular to apply eq 18 when $\beta \hbar \omega \ll 2\pi$ [and otherwise to abandon the parabolic model or to use eq 17 out of its range of validity], we see that eq 13 and 16 provide a physically and mathematically more satisfactory result that may be used for any value of $\beta \hbar \omega$.

C. Extension to Unsymmetric Barriers. The derivation in part A carries over to an exoergic barrier with no modification. For an endoergic barrier

$$V(x) = 0 \quad x < -(2V_0/m\omega^2)^{1/2}$$

$$V(x) = V_0 - 1/2 m \omega^2 x^2 \quad -(2V_0/m\omega^2)^{1/2} \leq x \leq [2(V_0 - V_1)/m\omega^2]^{1/2} \quad (20)$$

$$V(x) = V_1 \quad [2(V_0 - V_1)/m\omega^2]^{1/2} < x$$

where $V_1 < V_0$ and the particle is incident from the left,

TABLE I: Transmission Coefficients for Symmetric Truncated Parabolic Barriers

$V_0\alpha$	β/α	uni- formized WKB ^a	present algorithm ^b	untrun- cated para- bola ^c	Wigner meth- od ^d
1.0	0.10	0.98	0.97	1.02	1.02
1.0	0.50	1.03	0.96	1.57	1.41
1.0	0.90	1.24	1.01	9.15	2.33
1.0	1.05	1.36	1.08		2.81
1.0	1.65	2.10	2.32		5.48
1.0	2.45	4.16	5.51		10.87
2.0	0.10	1.00	1.00	1.02	1.02
2.0	0.50	1.22	1.20	1.57	1.41
2.0	0.90	1.87	1.78	9.15	2.33
2.0	1.05	2.28	2.21		2.81
2.0	1.65	5.76	6.78		5.48
3.0	0.10	1.01	1.01	1.02	1.02
3.0	0.50	1.35	1.35	1.57	1.41
3.0	0.90	2.91	2.48	9.15	2.33
3.0	1.05	3.37	3.40		2.81
3.0	1.65	13.65	15.30		5.48
4.0	1.05	4.59	4.65		2.81
4.0	1.65	29.52	31.64		5.48
5.0	0.10	1.02	1.02	1.02	1.02
5.0	0.50	1.47	1.47	1.57	1.41
5.0	0.90	3.69	3.69	9.15	2.33
6.0	1.05	7.26	7.35		2.81
6.0	1.65	120.17	122.87		5.48
9.0	0.10	1.02	1.02	1.02	1.02
9.0	0.50	1.56	1.57	1.57	1.41
9.0	0.90	3.49	3.49	9.15	2.33
9.5	1.05	12.68	12.77		2.81
9.5	1.65	1214	1217		5.48

^a Equations 4, 5, and 6. ^b Equation 21 with $V_2 = 0$.

^c Equation 18; results not shown for $\beta > \alpha$, i.e., for temperatures below the highest- T singularity of (18).

^d Equation 17 with ω fit to second derivative at top of barrier.

the only modification needed is to replace the lower integration limits in eq 2 and hence eq 9 by V_1 instead of zero. When this is done we obtain eq 11 but with $(V_0 - V_1)$ replacing V_0 . This gives the final result, generalizing (13) and (16):

$$\kappa(T) = \frac{\beta \pi / \alpha}{\sin(\beta \pi / \alpha)} - \frac{\beta}{\alpha - \beta} \exp[(\beta - \alpha)(V_0 - V_2)] \quad \beta \leq \alpha \quad (21a)$$

$$\kappa(T) = \frac{\beta}{\beta - \alpha} \{\exp[(\beta - \alpha)(V_0 - V_2)] - 1\} \quad \alpha \leq \beta$$

where

$$V_2 = \max(V_1, 0) \quad (21b)$$

The definition in eq 21b allows 21a to be applied to any truncated parabolic barrier (i.e., V_1 less than, equal to, or greater than zero) for any α and β .

III. Applications

A. Truncated Parabola. First we wish to show that eq 13 and 16 provide a numerically accurate alternative to numerical integration of eq 7 and 8. Table I shows a comparison of these results for various values of the unitless quantities $V_0\alpha$ and β/α . For $\alpha > \beta$ the results are also compared to the popular eq 18. The table shows that eq 18, which diverges at $\beta = \alpha$, already leads to serious errors at $\beta = 0.5\alpha$ for $V_0\alpha \leq 5$ and at $\beta = 0.9\alpha$ for $V_0\alpha = 9$. In contrast the new analytic formula, including the

(5) See, e.g., K. A. Quickert and D. J. LeRoy, *J. Chem. Phys.*, **52**, 856 (1970); W. H. Miller, *ibid.*, **61**, 1823 (1974).

TABLE II: Transmission Coefficients for Eckart Barriers^a

V_1, E_h	γ, a_0^{-1}	T, K	β/α_{top}	ω_{fit}, cm^{-1}	$\kappa(T)$			
					uniform semiclassical ^b	present algorithm ^c	untruncated parabola ^d	Wigner method ^j
-0.003	1.0 ^e	1500	0.071	465.9	1.008	1.008	1.008	1.008
		1000	0.107	465.8	1.019	1.019	1.019	1.019
		600	0.178	465.8	1.05	1.05	1.05	1.05
		400	0.267	465.5	1.13	1.13	1.13	1.12
		300	0.356	465.4	1.24	1.24	1.24	1.21
		200	0.535	464.9	1.66	1.68	1.69	1.47
		150	0.713	463.6	2.71	2.80	1.84	1.71
-0.003	3.0 ^f	1500	0.214	1390	1.08	1.08	1.08	1.08
		1000	0.321	1388	1.18	1.19	1.19	1.17
		600	0.534	1382	1.62	1.66	1.69	1.47
		400	0.802	1360	3.38	3.82	4.32	2.06
		300	1.069	1233	14.24	13.89		2.88
0.009	1.0 ^g	1500	0.063	414.5	1.007	1.007	1.007	1.007
		1000	0.095	414.5	1.015	1.105	1.015	1.015
		600	0.159	414.5	1.04	1.04	1.04	1.04
		400	0.238	414.3	1.10	1.10	1.10	1.09
		300	0.317	414.7	1.18	1.19	1.18	1.16
		200	0.476	413.7	1.48	1.49	1.49	1.37
		150	0.634	413.0	2.11	2.16	2.16	1.65
0.009	3.0 ^h	1500	0.190	1236	1.06	1.06	1.06	1.06
		1000	0.285	1235	1.14	1.14	1.15	1.13
		600	0.476	1231	1.45	1.48	1.50	1.37
		400	0.713	1218	2.47	2.69	2.81	1.83
		300	0.951	1165	6.28	7.70	17.21	2.48
0.009	5.0 ⁱ	200	1.427	894	438.4	878.9		4.30
		1500	0.317	2043	1.16	1.18	1.15	1.14
		1000	0.475	2033	1.42	1.47	1.39	1.31
		600	0.792	1978	2.87	3.32	2.94	1.85
		400	1.188	1633	16.15	13.70		2.92
		300	1.584	1491	324.4	721.5		4.41

^a $m = 3474$ au, $V_0 = 0.03 E_h$. See footnotes 8-10 for explanation of units. ^b Computed by uniformized WKB method using program of ref 11. ^c Equations 21 and 24-26. ^d Equation 18 with $\omega = \omega_{top}$: results not shown for $\beta > \alpha_{top}$, i.e., for temperatures below the highest- T singularity of (18). ^e $\omega_{top} = 465.6 cm^{-1}$. ^f $\omega_{top} = 1379 cm^{-1}$. ^g $\omega_{top} = 412.5 cm^{-1}$. ^h $\omega_{top} = 1237 cm^{-1}$. ⁱ $\omega_{top} = 1888 cm^{-1}$. ^j Equation 17 with $\omega = \omega_{top}$.

terms that cancel the divergences and lead to a continuous transition through the $\beta = \alpha$ region, provides useful accuracy even at $\beta = 1.65\alpha$. The most difficult regime for the new formula is the case of low barrier heights; thus the largest errors in the table are for $V_0\alpha = 1$. By $V_0\alpha = 3$, the new formula provides 1% accuracy up to $\beta = 1.05\alpha$ and by $V_0\alpha = 9$ it provides 0.3% accuracy up to $\beta = 1.65\alpha$.

B. Eckart Barrier. In this section we wish to show that eq 21 can also provide useful results for other, more realistic, barrier shapes. Consider the Eckart potential⁷

$$V(x) = \frac{V_1\gamma}{1+\gamma} + \frac{[V_0^{1/2} + (V_0 - V_1)^{1/2}]^2}{(1+\gamma)^2} \quad (22a)$$

where

$$\gamma = \exp(\gamma x) \quad (22b)$$

and γ is a range parameter. We wish to approximate this by eq 20 and then to use eq 21 to approximate $\kappa(T)$. The customary procedure is to use the second derivative at the barrier maximum to approximate ω and then to use eq 17 or 18 to approximate $\kappa(T)$. Our goal is to improve on both steps of the customary procedure.

First we estimate the "significant tunneling region" (ΔE_{STR} , a function of T). For this purpose it is sufficient to consider the un-uniformized WKB approximation $\exp(-\beta E) \exp[\alpha(E - V_0)]$ to the integrand of (5). We define ΔE_{STR} as the energy interval over which this factor drops

from $\exp(-\beta V_0)$ to $e^{-1} \exp(-\beta V_0)$ when the barrier of interest is fit to a parabola. This yields

$$\Delta E_{STR}(T) = 1/(\alpha - \beta) \quad (23)$$

where α is computed for the fit parabola. The energy interval within $\Delta E_{STR}(T)$ of the top of the barrier is the region that contributes most to $\kappa(T)$, and it is over this region $[(V_0 - \Delta E_{STR}) \leq E \leq V_0]$ that the parabolic fit must be best. At high T

$$\Delta E_{STR} \xrightarrow{T \rightarrow \infty} 1/\alpha$$

The second derivative at the barrier maximum seldom leads to a good fit over even this interval, and as T decreases, $\Delta E_{STR}(T)$ becomes even larger.

The general procedure we evolved to obtain a good effective parabolic fit to the true potential at a temperature T is as follows. Let $s = w - x_0$ where x_0 is the position of the maximum in the true $V(x)$. Consider a sequence of distances s_i where $s_1 = \delta$ and $s_{i+1} = s_i + \delta$. For each i ($i = 1, 2, \dots$), fit a parabola (as described in the Appendix) to the true potential over the range $(-s_i, +s_i)$. This yields ω_i and hence α_i . We define

$$\Delta E_i = V_0 - \frac{1}{2}[V(s_i) + V(-s_i)] \quad (24)$$

Increment i by 1 until either of the following conditions is satisfied:

$$\Delta E_i \geq \frac{1}{\alpha_i - \min(\alpha_{i\beta})} \quad (25)$$

or

$$\frac{1}{2}m\omega_i s_i^2 \geq \min[V_0, (V_0 - V_1)] \quad (26)$$

(6) See, e.g., H. S. Johnston, "Gas Phase Reaction Rate Theory", Ronald Press, New York, 1966.

(7) C. Eckart, *Phys. Rev.*, **35**, 1303 (1930).

TABLE III: Transmission Coefficients for Fit to Scaled SSMK Potential Energy Barrier^a

T, K	$\beta/\alpha_{\text{top}}$	$\omega_{\text{fit}}, \text{cm}^{-1}$	$\kappa(T)$				
			accurate quantal ^b	uniform semiclassical ^c	present algorithm ^d	untruncated parabola ^e	Wigner method ^f
1250	0.245	1303	1.14	1.09	1.10	1.11	1.10
1000	0.306	1301	1.21	1.14	1.16	1.17	1.15
600	0.510	1286	1.61	1.46	1.54	1.60	1.43
400	0.765	1245	2.79	2.42	2.83	3.57	1.96
350	0.874	1208	3.80	3.25	3.96	7.15	2.26
300	1.020	1131	6.21	5.26	6.04		2.71
250	1.224	999.0	14.74	12.45	9.17		3.46
200	1.530	854.5	83.48	71.97	19.02		4.85

^a $m = 1224.7333 \text{ au}$, $b_1 = 0.1129 \text{ eV}$, $b_2 = 0.2294 \text{ eV}$, $b_3 = 0.30855 \text{ a}_0^{-2}$, $b_4 = 1.11123 \text{ a}_0^{-2}$, $B = 0.424 \text{ eV}$, $\omega_{\text{top}} = 1337 \text{ cm}^{-1}$. ^b Reference 15. ^c Computed by uniformized WKB method using program of ref 11. ^d Equations 21 and 24-26. ^e Equation 18 with $\omega = \omega_{\text{top}}$; results not shown for $\beta > \alpha_{\text{top}}$. ^f Equation 17 with $\omega = \omega_{\text{top}}$.

The final value of ω_i yields the effective α to use in eq 21. These final values are called ω_{fit} and α_{fit} to distinguish them from ω_{top} and α_{top} which are calculated from the second derivative at the top of the barrier. We choose δ small enough (0.001 a_0) so that the final results are independent of decreasing it further. When $\alpha_1 > \beta$, eq 25 is a consequence of eq 23. When $\beta \geq \alpha_i$, the denominator of eq 25 ensures that it is not used. If $\beta \geq \alpha_i$, then tunneling through all portions of the barrier is important and eq 26, which reflects the finiteness of the barrier, must be used. For realistic potentials, if we did not enforce (26), as T decreases we would reach a point where the true potential at $E = V_0 - \Delta E_i$ is concave upward. The parabolic fit is always concave downward, and attempts to fit the true potential beyond this point lead to increasingly less realistic fits. Equation 26 forces ω_i to stop decreasing below the value at which a parabolic fit is still realistic.

To test this algorithm we performed calculations for a set of Eckart barriers. We set m equal to the mass of a deuterium atom⁸ and V_0 equal to $0.03 E_h$.⁹ We studied several different values for V_1 and γ . The parameter values¹⁰ and the results are given in Table II where they are compared to accurate uniform semiclassical results and to the results of the untruncated parabola formula and the Wigner method, i.e., using eq 17 or 18 with ω determined by the second derivative at the top of the barrier. The accurate uniform semiclassical results were calculated by the computer program KAPPAS¹¹ which is described elsewhere.¹² We see the present method provides reasonably accurate results for a wide range of parameters.

As another test we considered a least-squares fit to the potential energy barrier of Shavitt et al.¹³ (SSMK), scaled as suggested by Shavitt.¹⁴ This barrier is given by¹⁵

$$V(s) = b_1 \text{sech}^3(b_3 s^2) + b_2 \exp(-b_4 s^2) + (B - b_1 - b_2) \exp(-4b_4 s^2)$$

where the parameters are given in Table III. The results are given in Table III where they are compared to accurate quantal transmission coefficients¹⁵ and to the results obtained by applying the previous methods. Again we see that the new algorithm provides considerable improvement over previous parabolic tunneling formulas. Table III also

TABLE IV: Transmission Coefficients for Gaussian Barrier^a

T, K	$\omega_{\text{fit}}, \text{cm}^{-1}$	uniform semi- classical	present algo- rithm ^c	untrun- cated para- bola ^d	Wigner method ^e
1250	1321	1.10	1.10	1.11	1.10
1000	1319	1.16	1.17	1.17	1.15
600	1312	1.52	1.57	1.60	1.43
400	1289	2.78	3.13	3.57	1.96
350	1262	4.05	4.75	7.15	2.26
300	1186	7.77	8.00		2.71
250	1102	27.61	21.65		3.47
200	1102	433.75	791.3		4.85

^a Barrier height = $0.0155819 E_h$, $m = 1224.7333 \text{ au}$, $\omega_{\text{top}} = 1337 \text{ cm}^{-1}$. ^b Computed by uniformized WKB method using program of ref 11. ^c Equations 21 and 24-26. ^d Equation 18 with $\omega = \omega_{\text{top}}$; results not shown for $\beta > \alpha_{\text{top}}$. ^e Equation 17 with $\omega = \omega_{\text{top}}$.

shows, as already discussed elsewhere for other cases,^{12,16,17} that the uniformized WKB results are in reasonably good agreement with accurate quantal results for the one-mathematical-dimensional barrier transmission problem.

A final example is provided in Table IV where we consider a Gaussian barrier with the same height and ω_{top} as the barrier in Table III. For this shape of barrier too, the present procedures provide useful accuracy.

IV. Concluding Remarks

The main result of this paper is eq 21, which is a continuous, divergence-free analytic expression that gives reasonable results for the Boltzmann-averaged transmission coefficient for a truncated parabolic barrier. We showed that this expression may also be used to provide simple analytic approximations for other shapes of barriers. A popular approach in the literature is to use the parabolic expression (17) to approximate the transmission coefficient for nonparabolic barriers in terms of the second derivative at the barrier maximum (which yields ω). Although the conditions for validity of this formula are seldom satisfied for problems of chemical interest,¹⁸ it does often provide results of useful empirical accuracy.¹⁹ We have shown that a much more satisfactory parabolic approximation can be based on the present formula (21) and the effective par-

(8) 3474 atomic units (au) of mass.

(9) $1 E_h = 1 \text{ au of energy} = 1 \text{ hartree}$. Therefore, $0.03 E_h = 0.81635 \text{ eV} = 18.825 \text{ kcal/mol}$.

(10) $1 a_0 = 1 \text{ au of length} = 1 \text{ bohr radius} = 0.52918 \times 10^{-10} \text{ m}$.

(11) B. C. Garrett and D. G. Truhlar, "National Resource for Computation in Chemistry Software Catalog", Vol. 1, Lawrence Berkeley Laboratory, Berkeley, CA, 1980, Program No. KS01(KAPPAS).

(12) B. C. Garrett and D. G. Truhlar, *J. Phys. Chem.*, **83**, 2921 (1979).

(13) I. Shavitt, R. M. Stevens, F. L. Minn, and M. Karplus, *J. Chem. Phys.*, **48**, 2700 (1968).

(14) I. Shavitt, *J. Chem. Phys.*, **49**, 4048 (1968).

(15) D. G. Truhlar and A. Kuppermann, *J. Am. Chem. Soc.*, **93**, 1840 (1971).

(16) B. C. Garrett, D. G. Truhlar, R. S. Grev, and A. W. Magnuson, *J. Phys. Chem.*, **84**, 1730 (1980); B. C. Garrett, D. G. Truhlar, R. S. Grev, and R. B. Walker, *J. Chem. Phys.*, **73**, 235 (1980).

(17) B. C. Garrett, D. G. Truhlar, and R. S. Grev, *J. Phys. Chem.*, **84**, 1749 (1980); B. C. Garrett, D. G. Truhlar, R. S. Grev, A. W. Magnuson, and J. N. L. Connor, *J. Chem. Phys.*, **73**, 1721 (1980).

(18) P. Pechukas in "Dynamics of Molecular Collisions", Part B, W. H. Miller, Ed., Plenum Press, New York, 1976, p 269.

(19) B. C. Garrett and D. G. Truhlar, *J. Phys. Chem.*, **83**, 200, 3058(E) (1979); **83**, 1079 (1979), **84**, 682(E) (1980).

TABLE A-I: Sensitivity of the Results in Table III to Fitting Procedure

T, K	$\kappa(T)^a$			
	uniform semi- classical ^b	$0, \pm s_i/2$	$0, \pm 3s_i/4$	$0, s_i/2, \pm s_i$
1250	1.09	1.10	1.10	1.10
1000	1.14	1.17	1.16	1.16
600	1.46	1.57	1.54	1.54
400	2.42	3.11	2.83	2.81
350	3.25	4.63	3.96	3.93
300	5.26	7.38	6.04	6.06
250	12.46	31.85	9.17	9.40
200	71.97	1416	19.02	54.84

^a Equation 21 with fitting procedures of Appendix 1.^b Computed by uniformized WKB method using program of ref. 11.

abolic width of the nonparabolic barrier at the energies that contribute appreciably to the transmission coefficient. This allows the parabolic formulas to be used in a more meaningful way. In another paper²⁰ we shall show that the parabolic approximation discussed here can be combined with a simple but realistic treatment of reaction-path curvature to provide a nonempirical analytic formula for the transmission coefficient of atom-diatom reactions that rivals in accuracy the best state-of-the-art semiclassical treatments^{16,21} for H + H₂ and isotopic analogues. Since it is much easier to approximate a reaction barrier para-

bolically than to calculate it fully, the expression proposed in this paper may be very useful for practical work.

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Appendix 1

To fit a parabola to a potential over the interval $(-s_i, +s_i)$ where $s = 0$ denotes the maximum of the true potential, we first equate a cubic polynomial to the potential at $0, \pm 3s_i/4$ (this was found by trial to give better results than $0, \pm s_i/2$ or $0, \pm s_i$). The cubic polynomial is written as $V_{\text{cubic}} = V_0 + \frac{1}{2}m\omega_i^2 s^2 + cs^3$ since the first derivative is zero at $s = 0$. The effective parabolic fit is obtained by setting $c = 0$. The procedure just described was used for all the results in Tables II and III; however, one can sometimes obtain better results by more complicated procedures. For example, we also tried fitting cubic polynomials to the potential at two sets of points $0, \pm s_i$ and $0, \pm s_i/2$ and defining the best parabolic fit by averaging the quadratic coefficients obtained in the two fits. Although this five-point fit sometimes gives better results, the extra complication does not seem warranted.

Table A-I shows the sensitivity of the present results to the fitting procedure for the case shown in Table III. As usual, the $0, \pm 3s_i/4$ fit works better than the other three-point fits. This case is less typical though as regards the five-point fit; this is the case where the five-point fit makes the most difference as compared to the $0, \pm 3s_i/4$ fit and where it offers the most improvement.

(20) R. T. Skodje, D. G. Truhlar, and B. C. Garrett, to be published.

(21) R. A. Marcus and M. E. Coltrin, *J. Chem. Phys.*, **67**, 2609 (1977); W. H. Miller, N. C. Handy, and J. E. Adams, *ibid.*, **72**, 99 (1980).